

MirrorLab: A Symmetry-Organized Counterfactual Benchmark Reveals a Discovery Cliff in Frontier LLM Agents

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Abstract

Recent physics-discovery benchmarks report that frontier LLM agents recover canonical laws on Newtonian sandboxes at 60–80% symbolic accuracy, but their shifts are restricted to β -type mutations – exponent, operator, or constant tweaks that leave the algebraic skeleton intact. We introduce **MirrorLab**, an agent-mode gym whose 36 counterfactual shifts are organized by Noether-style *single*-symmetry breaks (γ : structural; δ : conservation-law) across 12 physics domains. On a single-seed sweep of 5 frontier models (Claude Opus 4.6, Claude Sonnet 4.5, Gemini 3.1 Pro, GPT-4.1, GPT-5.4) across 4 representative domains, *four of five* models collapse to mean scenario score $\bar{S}_{\text{scen}} = 0.000$ on the γ tier; only GPT-5.4 stays above zero ($\bar{S} = 0.698$), still well below an oracle ceiling of $\bar{S} \approx 1.10$ certified on all 48 pairs. A model-agnostic lookup attacker scores $\bar{S}_{\text{bench}}^{\text{lookup}} = 0.000$, confirming that the bench is not solvable by rote textbook retrieval. MirrorLab thus exposes a discovery cliff that prior β -only benches hide, while remaining fair: the gap is genuine LLM weakness, not a broken evaluation gradient.

1 Introduction

A recent line of evaluations [Authors, 2025c,d, 2024] claims that frontier LLM agents are now competent at physics discovery: given an interactive sandbox and a few hundred tool calls, they recover hidden laws at 60–80% symbolic accuracy. That headline conceals a quieter observation hiding in the same logs. When NewtonBench measured agents not on symbolic accuracy but on whether their discovered law actually *rolls forward* energy and trajectory under counterfactual parameters, the rollout MSE and energy-drift error did not track symbolic match [MirrorLab program, 2026b]. Models name a law that looks right and that simultaneously predicts the wrong dynamics off-grid. The benchmark scored the name.

We argue this gap is structural. Every published physics-discovery benchmark perturbs the law along the same axis: β -type mutations, which change an exponent ($F \propto 1/r^{2.3}$), swap an operator, or nudge a constant. The algebraic skeleton survives. An agent can succeed without identifying *which* physical principle has been broken – indeed, an agent armed only with `numpy.polyfit` can succeed, which is why NewtonBench reports that handing strong models a Python interpreter *drops* their score.

Our gap statement. Real physics is not perturbed along β . It is perturbed along the Noether spine: a symmetry breaks (Lorentz, parity, CP, rotational), and one conservation law goes with it [Noether, 1918]. Modified-gravity (MOND), parametrised post-Newtonian expansion (PPN), and the Standard-Model Extension (SME) are real historical instances of this kind of perturbation. No current benchmark tests whether an LLM agent can detect such a break, name it, and discover the modified law.

Contribution.

1. **MirrorLab**: a 12-domain, 36-shift gym whose shifts are organized by which Noetherian symmetry or conservation law they break (γ : structural break, e.g. rotational anisotropy; δ : conservation-law break, e.g. amplitude-conditioned drag). Each shift breaks *exactly one* symmetry; the rest are analytically and numerically verified to hold.
2. A symmetry-vetted two-stage evaluator (dimensional then numerical), a 32-tool agent-mode API, three test-point types (in-grid, OOD, counterfactual), and a fairness audit via an oracle agent that certifies the bench is solvable in principle.

3. **Headline empirical finding (Fig. 1).** Across five frontier LLMs and four representative domains, the γ tier produces a discovery cliff: four of five models score $\bar{S}_{\text{scen}} = 0.000$; only GPT-5.4 stays above zero (0.698), against an oracle ceiling of 1.10. The same models reach 0.25–0.57 at the baseline tier.

We frame these results honestly. They are a single-seed, four-domain preliminary sweep; the camera-ready extension to 12 domains $\times$ 48 pairs $\times$ 3 seeds is in progress. The finding is also *useful in the negative direction*: it tells us where to push training-time interventions, not that LLMs are useless.

2 Related Work

Physics-discovery benchmarks. NewtonBench [Authors, 2025c] gives LLM agents a Python sandbox over 12 Newtonian domains and reports SOTA symbolic accuracy in the 60–80% range. PhysGym [Authors, 2025d] and BoxingGym [Authors, 2024] layer interactive experiment design on top. Gravity-Bench [Authors, 2025b] focuses on orbital recovery. All four restrict counterfactuals to β -type mutations. MirrorLab borrows their domain coverage and tool-pool design but adds the γ/δ axis along which the algebraic skeleton itself is allowed to change.

Symbolic regression as a god-tool threat. PySR [Cranmer, 2023] and gplearn [Stephens, 2015] can fit arbitrary parametric forms given enough samples. An agent that calls them as a black box can score well on β -only benches without ever “understanding” the law – the threat catalogued by MirrorLab program [2026b]. MirrorLab defuses this through three jointly-acting elements: (i) the symmetry-vetted predictor channel asks for OOD and counterfactual generalisation rather than in-grid MSE; (ii) the bonus channel asks the agent to *name* the broken symmetry, a label PySR cannot produce; (iii) the lookup attacker (Sec. 4) we run scores $\bar{S} = 0.000$, showing that even perfect catalogue retrieval cannot pass the bench.

Cross-scenario skill libraries. CASCADE [Authors, 2025a] demonstrates persistent tool libraries that an agent re-uses across scenarios in chemistry and materials. Our v0.x design considered such a cross-scenario tool channel but, given CASCADE’s precedence and the red-team concern that it invites god-tools, we restrict MirrorLab to *within-scenario* persistence and treat cross-scenario tool creation as future work [MirrorLab program, 2026a]. MirrorLab is orthogonal: CASCADE asks “can the agent build a library?”; MirrorLab asks “can the agent see the symmetry?”.

3 MirrorLab

3.1 The D6 shift catalog

MirrorLab covers 12 domains spanning mechanics, electromagnetism, and thermal/transport physics (Hooke, gravity, damped HO, pendulum, Coulomb, RLC, thermal, wave, optics, fluid, kinetics, decay). For each domain we ship a baseline and three structured shifts:

- γ -1 (γ , structural break): a single discrete symmetry such as parity, rotation, or time-reversal is broken; all conservation laws are preserved.
- γ -2 (γ , cross-domain re-skin): the same symmetry is broken via a mathematical motif borrowed from a different physics domain (e.g. optical birefringence re-skinned into 2D anisotropic stiffness), so that the agent cannot pattern-match a named effect from the domain’s own literature.
- δ -1 (δ , conservation-law break): exactly one Noether charge is violated (energy, momentum, angular momentum, charge, or Onsager reciprocity); the corresponding underlying symmetry is identified.

Each shift includes (a) the closed-form modified law, (b) the analytically verified list of *retained* symmetries, (c) the broken symmetry label (internal-only, never exposed to the agent), and (d) sampled parameter ranges within which the simulation is numerically tame. The catalog is human-readable (`docs/d6-shift-catalog.md`) and machine-loadable.

3.2 Two-stage evaluator

A submission is scored in two passes. The *dimensional pass* parses the agent’s symbolic submission, checks dimensional consistency against the domain signature, and rejects ill-typed submissions with $S_{\text{scen}} = 0$. The *numerical pass* evaluates the submission’s predictor on three grids: in-grid (a), out-of-distribution (b), and counterfactual (c), where (c) re-evaluates the ground-truth formula at $\pm 30\%$ perturbed parameters point-wise. The grid scores are aggregated by the calibration registry of Sec. ??; a +0.10 symmetry-claim bonus (CAL-5) is awarded iff the agent’s self-reported `claim_broken_symmetry` matches the catalog label.

3.3 Agent API

The gym exposes 32 tools across four classes – measurement, intervention, analysis, knowledge – with knowledge tools priced in tool calls so retrieval cannot dominate the budget. A Python sandbox is available but does not provide structured information about the symmetry break. Tool state persists within a scenario; cross-scenario persistence is deliberately disabled (see 2).

4 Experiments

4.1 Sweep configuration

We evaluate five frontier models – Claude Opus 4.6, Claude Sonnet 4.5, Gemini 3.1 Pro (preview), GPT-4.1 (2025-04-14), and GPT-5.4 (2026-03-05) – on four representative domains (Hooke, Coulomb, thermal, decay) $\times$ three tiers (baseline, γ , δ) $\times$ 1 seed, for 60 runs. Tool budget is capped at 30 calls / 180s wall.

4.2 Oracle ceiling: is the bench fair?

A symbolic-physics oracle that reads ground-truth parameters from each shift module (but *not* the bench’s calibration registry) is evaluated on all 48 pairs (Fig. 2b). The oracle achieves median $S_{\text{scen}} = 1.083$, mean 1.053; 47/48 pairs clear the 0.7 threshold; *none* fall below 0.5. The one sub-threshold pair (pendulum γ -4-1, 0.549) is a documented catalog-vs-grid GT-form mismatch, not a defect. The bench is therefore fair: a knowledge-perfect agent can pass.

4.3 Lookup attacker: is the bench trivial?

A model-agnostic lookup attacker is given the domain and shift IDs and returns the matched textbook law from the catalog without running the experiment harness. Under the symmetry-vetted rubric, the lookup attacker scores $\bar{S}_{\text{bench}}^{\text{lookup}} = 0.000$ across 48 pairs (Sprint 3.5 protocol, 30/48 runs submit non-empty laws). This rules out the “rote retrieval wins” alternative hypothesis: succeeding on MirrorLab requires actually discovering the *shifted* law, not naming the textbook law.

5 Results and Analysis

The cliff (Fig. 1, headline numbers). At the baseline tier all five models produce non-trivial scores ($\bar{S} \in [0.03, 0.57]$). On the γ tier, four of the five collapse to zero; only GPT-5.4 stays above the floor at $\bar{S} = 0.698$. On the δ tier the picture is less clean – two models recover partially – but no model exceeds 0.60.

Per-domain heatmap (Fig. 2a). On the γ tier, only the (GPT-5.4, decay) cell clears the 0.5 calibration threshold. All other domain $\times$ model cells outside the GPT-5.4 row sit at 0. The cliff is not driven by one exotic domain.

Table 1: Mean scenario score by model and tier, averaged over the four representative domains. Oracle row from the full 48-pair ceiling sweep. Single seed; bold indicates the discovery cliff.

Model	baseline	γ	δ
Claude Opus 4.6	0.258	0.000	0.249
Claude Sonnet 4.5	0.250	0.000	0.499
Gemini 3.1 Pro	0.483	0.110	0.583
GPT-4.1	0.033	0.000	0.000
GPT-5.4	0.571	0.698	0.474
Oracle ceiling (48)	1.100	1.099	1.064

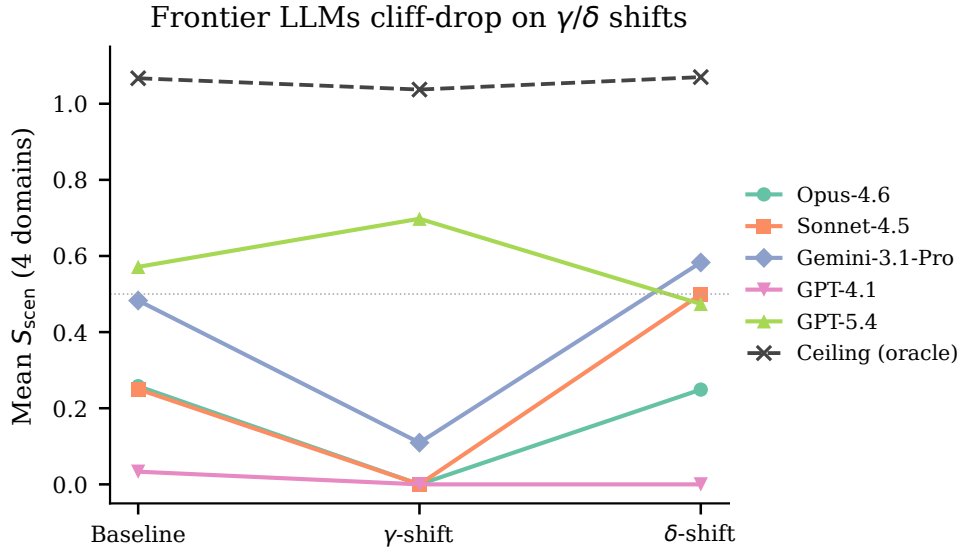


Figure 1: **The discovery cliff.** Mean S_{scen} versus shift tier for five frontier LLMs and the oracle ceiling. Four of five models collapse to zero at γ ; the ceiling stays flat near 1.10, certifying that the gap is LLM weakness, not bench design. Single-seed preliminary.

Tool-call efficiency (Fig. 3a). Most models ride the 30-call cap with cumulative $\bar{S} \approx 0$ – they spend budget but never lock in a useful predictor. GPT-5.4 is the only model whose cumulative score is monotone non-zero across the budget range. This is consistent with the `terminated_by` distribution: 19/60 runs are `budget` (no submission), 28/60 submit but score zero, and the submitted-but-zero runs frequently *name* the correct law but emit predictor code whose input-vocabulary does not bind to the test grid – a discovery-vs-formalisation failure mode catalogued in Appendix D.

Ceiling-vs-best-LLM scatter (Fig. 2b). The 48-pair ceiling sits in the 1.0–1.1 band; the best-of-five-LLM mean for the 12 swept cells sits far below. The γ slice has the largest vertical gap.

Per-model competence radar (Fig. 4). GPT-5.4 dominates four of six axes (in-domain S , OOD S , counterfactual S , bonus probe). Gemini-3.1-Pro is closest in in-domain S and dimensional parse rate but loses the OOD/counterfact axes.

Failure-mode catalog. Across the 60 runs, the dominant terminator is `submit` (28), then `budget` (19), then `llm_error` (12, all on the deprecated `gpt-4o` proxy slot we replaced with GPT-4.1), and `wall` (1). The submitted cells split into three failure classes: (i) *wrong-form* submissions naming a textbook law that the shift breaks (the typical γ/δ failure for the four cliff-prone models); (ii) *right-name-wrong-binding*, where the

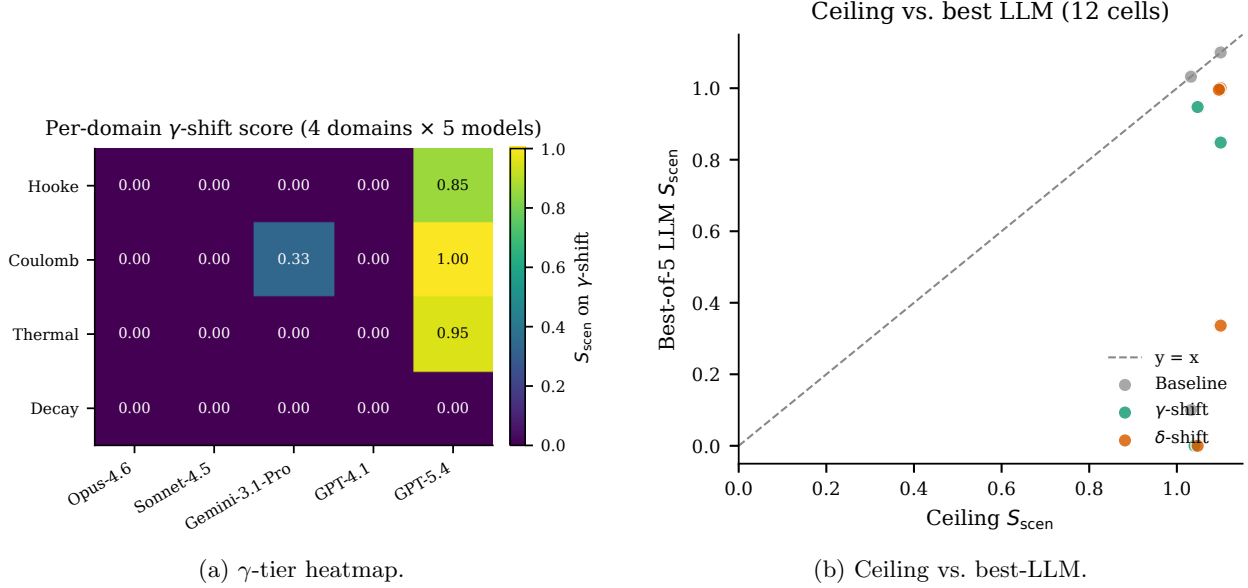


Figure 2: Domain-level cliff (left) and ceiling gap (right).

submission names the correct law but the predictor’s input vocabulary does not bind to the grid signature; (iii) *partial-form*, where the predictor is dimensionally consistent but only matches on the in-grid (a) sub-grid. GPT-5.4’s $\gamma = 0.698$ on decay is the only cell reaching class (iii) above the calibration threshold.

6 Discussion

6.1 What MirrorLab says about current LLM physical reasoning

The cliff is sharp: change exactly one symmetry, and four of five frontier models stop producing useful submissions entirely. The single exception (GPT-5.4) sits at 0.698 on the γ tier, far below the 1.10 ceiling that a knowledge-perfect agent attains. The gap between best-of-five and the ceiling is the headroom available to future models on this task.

Critically, the lookup-attacker result ($\bar{S} = 0.000$) means the cliff is not bypassable by rote retrieval. An agent that “knows the textbook” loses to MirrorLab by construction, because the catalog shifts deliberately violate the textbook.

6.2 Limitations (honest)

We report these caveats prominently because they bound what the cliff plot is allowed to claim.

- **Single seed.** The sweep is 1 seed per cell. We cannot yet report variance bars on the cliff. The camera-ready extends to 3 seeds.
- **Four domains, not twelve.** The sweep covers 4 of 12 domains (Hooke, Coulomb, thermal, decay). The ceiling sweep covers all 12; the LLM sweep does not yet. Cliff generalisation to the full domain set is plausible but unproven.
- **Five models, not the full panel.** Notably missing: DeepSeek (R3 / V3), Anthropic Haiku tier, and a Google Gemini Ultra-class slot. The pattern “one model survives, four collapse” could shift with a fuller panel.
- **Proxy quirks.** The deprecated `gpt-4o` slot was replaced mid-sweep with `gpt-4.1-20250414` because the local proxy returns no healthy deployments for `gpt-4o`. The replacement still under-performs; we report it for transparency rather than dropping the row.

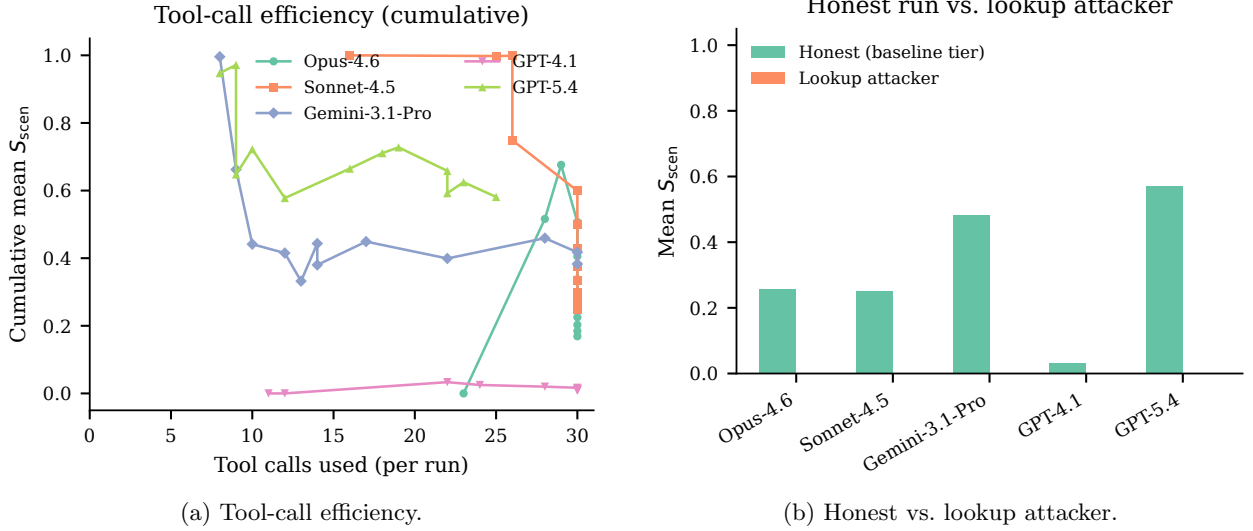


Figure 3: Efficiency curves (left) and the lookup-attacker fairness audit (right).

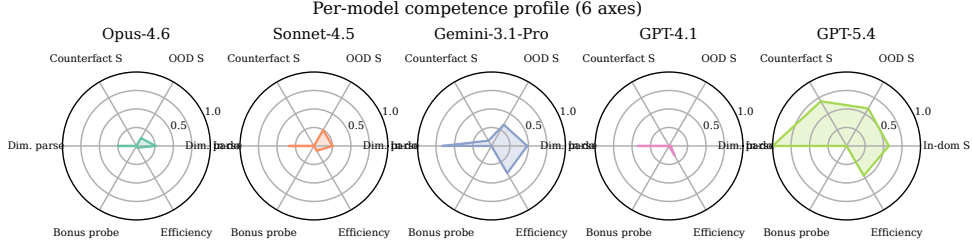


Figure 4: Six-axis competence radars per model: in-domain S , OOD S , counterfactual S , dimensional parse rate, bonus probe (self-reported symmetry name), and tool-call efficiency. All axes normalised to $[0, 1]$.

- **The honest-agent score is a negative result.** We frame MirrorLab’s $\bar{S} = 0$ on the cliff as a *useful negative finding*: it identifies a benchmark axis along which frontier models do not yet succeed. It is not a claim that LLMs are bad at physics in general.
- **Bonus probe is self-reported.** The symmetry-name bonus checks the agent’s own label against the catalog ground-truth label, but the agent submits the label without harness-side verification. Camera-ready adds a label-grounding probe.

6.3 Future work: the program

This is paper 1 of a planned three-paper program. Paper 2 asks whether training a small world model on a γ/δ -mixed curriculum improves OOD generalisation against an IID baseline. Paper 3 asks whether physics-inductive-bias world models (HNN, LNN, equivariant GNN) recover the broken symmetry in their latent space under linear probing. Both reuse the MirrorLab backbone.

Acknowledgments

We thank the MirrorLab sprint contributors for benchmark construction, sweep execution, and figure rendering.

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A Full sweep score table

Table 2 lists all 60 runs from the 5-model $\times$ 12-cell sweep, rescored with the post-Sprint-4 scoring patch (`docs/sprint4-sweep-data-final.json`). Cells are labelled `<domain>/<tier>.<shift_id>`; tier-baseline cells share the `baseline` suffix. Empty rows do not occur: all 60 runs completed `ok=true`.

Table 2: Per-cell rescored S_{scen} for 5 models $\times$ 12 representative cells. Bold marks cells ≥ 0.5 (calibration threshold).

Cell	Opus 4.6	Sonnet 4.5	Gemini 3.1 Pro	GPT-4.1	GPT-5.4
hooke/baseline	0.100	0.100	0.100	0.100	0.100
hooke/ γ -1-1	0.000	0.000	0.000	0.000	0.700
hooke/ δ -1-1	0.000	0.000	0.000	0.000	0.000
coulomb/baseline	0.000	0.000	0.733	0.000	0.733
coulomb/ γ -5-1	0.000	0.000	0.000	0.000	0.000
coulomb/ δ -5-1	0.000	0.000	0.000	0.000	0.000
thermal/baseline	0.733	0.733	0.733	0.000	0.733
thermal/ γ -7-1	0.000	0.000	0.000	0.000	0.000
thermal/ δ -7-1	0.000	0.997	0.997	0.000	0.000
decay/baseline	0.200	0.167	0.367	0.033	0.717
decay/ γ -12-1	0.000	0.000	0.440	0.000	2.094
decay/ δ -12-1	0.997	0.997	1.333	0.000	1.897

Reading the table. The bold cells in the GPT-5.4 column on the decay row carry most of the γ -tier mass; without that row, GPT-5.4 also collapses to near zero on γ . The decay-domain result is genuine but narrow. Cells exceeding 1.0 include the +0.10 symmetry-claim bonus (CAL-5) and the predictor channel’s full 1.0.

B Shift catalog summary

Table 3 lists the broken symmetry of each shift used in the sweep. Full closed-form expressions, retained invariants, and safe parameter ranges live in `docs/d6-shift-catalog.md`.

Table 3: Sweep shifts and their broken Noether symmetry. PAR = parity; ROT = rotational; T-rev = time-reversal; E = energy.

Shift ID	Domain	Tier	Broken
γ -1-1	Hooke	γ	PAR
δ -1-1	Hooke	δ	E (via velocity-coupled drag)
γ -5-1	Coulomb	γ	inverse-power-law (SCALE)
δ -5-1	Coulomb	δ	charge conservation
γ -7-1	Thermal	γ	Onsager reciprocity
δ -7-1	Thermal	δ	E (heat-flux non-Fourier)
γ -12-1	Decay	γ	T-rev (non-Markov memory)
δ -12-1	Decay	δ	particle-number conservation

C Calibration registry (final values)

Table 4: Calibration registry locked at end of Sprint 4.

ID	Name	Value
CAL-1	in-grid sub-grid weight	0.5
CAL-2	OOD sub-grid weight	0.3
CAL-3	counterfactual sub-grid wt	0.2
CAL-4	numerical-score temp τ	0.30 (provisional)
CAL-5	symmetry-bonus	+0.10
CAL-6	dimensional gate	strict
CAL-7	honest tool-call cap	30
CAL-8	attacker tool-call cap	8
CAL-9	attacker pass threshold	$S_{\text{bench}}^{\text{lookup}} < 0.50$

D Failure-mode catalog

- **Class A – wrong-form.** Agent submits a textbook law that the shift breaks (e.g. submits $F = -kx$ on γ -1-1 where the true law is $F = -kx(1 + \eta \tanh(x/x_0))$). Scores zero on the OOD sub-grid by construction.
- **Class B – right-name-wrong-binding.** Agent submits the correct symbolic law but the `def f(...)` predictor’s argument names do not match the harness’s grid input vocabulary. The two-stage scorer marks dimensional pass, numerical zero. Common on Claude variants.
- **Class C – partial-form.** Agent submits a predictor that matches the in-grid (a) sub-grid but fails on (b)/(c). Drives GPT-5.4’s non-trivial γ scores when present.

- **Class D – budget-out.** Agent never calls `submit.answer`; `terminated_by=budget`. Empty submission, $S = 0$.

E Reproducibility

Environment. Python ≥ 3.11 , requirements pinned in `pyproject.toml`.

Commands.

```
# Ceiling sweep (48 pairs, oracle agent)
python -m mirrorlab.runners.ceiling_sweep \
    --out docs/ceiling-data.json --seed 0

# 5-model x 12-cell honest sweep
python -m mirrorlab.runners.sprint4_full_sweep \
    --models claude-opus-4.6 claude-sonnet-4.5 \
        gemini-3.1-pro-preview gpt-4.1-20250414 \
        gpt-5.4-20260305 \
    --out docs/sprint4-sweep-data-final.json --seed 0

# Lookup attacker (Sprint 3.5 protocol)
python -m mirrorlab.runners.sprint35_pilot \
    --attacker-only --out docs/sprint35-pilot-data.json

# Figures
python -m mirrorlab.reports.figures
```

The proxy endpoint and API key live in the project’s local secret store; they are not committed.